Supplemental Materials for "FM-CrossTraining: Computational Framework and Open Benchmark for Multiskill Maintenance Dispatching"

Ziheng Zhang and Tzu Pei Ku Chia

This document contains Supplemental Texts S1–S8, Tables S1–S22, and Figs. S1–S6, cited from the main manuscript. Symbols follow the Notation list of the main manuscript unless defined here.

Contents

S1 Overlay Construction, Format, and Unit Tests **S2**

S2 Admissible Reward Bound: Statement, Proof, and Verification **S3**

S3 Secondary Metrics and Latency Accounting **S4**

S4 Locked Defaults: Hyperparameters, Seeds, and Budgets **S5**

S5 Statistical Detail **S7**

S6 Data Contract, Reuse Boundary, and Descriptives **S10**

S7 Main-Text Exhibits in Full **S12**

S8 Sensitivity Analyses, Hypothetical Crew Sizes, and Matched Aggregate Load **S18**

S1 Overlay Construction, Format, and Unit Tests

This text expands the overlay construction summarized in the Framework section of the main manuscript. A resource overlay replaces the technician roster of a released work-order instance with a skill-structured roster at unchanged headcount, so the workload is held byte-identical while the skill structure is swept as a designed input. Table S1 lists the verdict campuses, their headcounts, and the realized skill-membership budget B that each rung of the flexibility ladder produces on them.

Table S1. Skill structures and realized skill-membership budgets B per verdict campus (m = 1.0). $n = 4$ campuses; full table for all six campuses in the repository.

Campus	Headcount	Trades K	B(CHAIN(1.0))	B(GEN)	B(FULL)
5	116	11	116	120	1160
9	99	13	99	96	1188
10	154	13	154	156	1848
12	186	13	186	192	2232

Overlay Format

An overlay is a JSON object naming the campus, the structure label Λ, the adoption fraction φ (chains only), the efficiency η, the crew multiplier m, the chain order, the realized skill-membership budget B, and the full technician list, each entry with a primary trade and a skill set. This budget counts added skill memberships, $B = \sum_u (|S_u| - 1)$. It is not a monetary training budget: the Design Curve section of the main manuscript prices structures under three cost accountings. A configuration is a pair (single-skill instance, overlay). Overlays are deterministic functions of the released crew calibration (the p95-weekly-hours capacity table), the structure parameters, and a recorded construction seed, so any third party can regenerate them bit-for-bit. They are also shipped as files.

Construction

Table S2 lists the pseudocode that constructs every overlay deterministically from the released capacity table. Crew scaling comes first: each trade's crew multiplier gives $c'_k = \max(1, \mathrm{round}(c_k m))$, following the calibration convention. Technicians are then numbered sequentially over the trades in sorted name order, and skills are assigned to the scaled roster. The chain order sorts trades by descending p95 weekly labor hours, the statistic the crews were calibrated on, and breaks ties by trade name. The three sensitivity permutations replace this order with seeded shuffles, whose seeds are recorded in the protocol log. The generalist pool distributes $n_{\mathrm{gen}} = \mathrm{round}(N/(K-1))$ full generalists across the trades in proportion to crew size, apportioned by largest remainder, with ties broken by larger crew and then by trade name. Within each trade, the lowest-indexed technicians are converted to generalists.

Unit Tests

Unit tests cover budget accounting, chain completeness (the $\varphi = 1$ eligibility graph is one cycle covering all trades), L0 equality with the single-skill workforce, crew-multiplier composition, and the degenerate one- and two-trade topologies. Those degenerate topologies behave sensibly and are tested explicitly: a single-trade campus collapses every structure to L0, and a two-trade campus makes CHAIN(1.0) coincide with FULL.

Table S2. Overlay construction pseudocode.

Input: campus crews $\{(g_k, c_k, v_k)\}$, structure, φ, η, m, permutation seed (optional)
1: for each trade k: $c'_k \leftarrow \max(1, \text{round}(c_k m))$
2: number technicians T0, T1, ... over trades sorted by name; $S_u \leftarrow \{\text{prim}(u)\}$
3: chain order $\leftarrow$ trades sorted by descending v_k (tie: name), or the seeded permutation
4: if CHAIN(φ): for chain position k, add successor skill to the $\lceil \varphi c'_k \rceil$ lowest-indexed technicians of trade t_k (no-op when $K = 1$)
5: if GEN: convert the apportioned n_{gen} lowest-indexed technicians per trade to $S_u = G$
6: if FULL: $S_u \leftarrow G$ for all u
7: record chain order, realized budget $B = \sum_u (|S_u| - 1)$, provenance tags, seeds

S2 Admissible Reward Bound: Statement, Proof, and Verification

This text gives the full statement, proof, and verification of the admissible reward bound for overlapping skill pools. The Methods Under Test section of the main manuscript states the same formulation as numbered equations, and the symbols below match it.

The potential-shaped reward uses $\Phi(s) = \text{realized}(s) + \text{LB}(s)$, where $\text{LB}(s)$ must never exceed the weighted tardiness that the queued jobs will still incur. The single-skill reward decomposed LB per trade. Skill overlap breaks that decomposition, so we derive the bound below for overlapping skill pools.

Per-job bound. For queued job j let $\tau_{\min}(j) = \min_{u \in M_j} \max(t, \tau_u)$, with τ_u the time technician u next becomes free. Every schedule satisfies $C_j \geq \tau_{\min}(j) + p_j$. The job cannot start before an eligible technician is available, and its realized duration is at least p_j because $\eta \leq 1$ only inflates durations. Hence, for any job set A,

$$\text{LB}_{\text{i}}(A) = \sum_{j \in A} w_j \left(\tau_{\min}(j) + p_j - d_j\right)^+ \tag{1}$$

is admissible.

Group overflow bound. Fix a trade subset $T' \subseteq G$ and a deadline d, and let $A(T', d)$ be the queued jobs with $g_j \in T'$ and $d_j \leq d$. Their nominal work is $D = \sum_{j \in A} p_j$, a lower bound on the realized work. The machine time available to the group by d is at most $\text{cap}(T', d) = \sum_{u: S_u \cap T' \neq \emptyset} (d - \tau_u)^+$, a deliberate over-count that preserves validity. With overflow $O = (D - \text{cap})^+$ and $k' = |\{u : S_u \cap T' \neq \emptyset\}|$ servers, the fluid argument of the single-skill bound gives a remaining-work area of at least $O^2/(2k')$ past d. Converting area to objective requires a constant that lower-bounds $w_j/\tilde{p}_j$ over every realized duration $\tilde{p}_j$ the job can take. Since $\tilde{p}_j \in \{p_j, \lceil p_j/\eta \rceil_{0.01}\}$ and the grid rounds up, the exactly admissible constant is

$$\kappa(T', d) = \min_{j \in A(T', d)} \frac{w_j}{\lceil p_j/\eta \rceil_{0.01}}, \qquad \text{LB}_{\text{ii}}(T', d) = \kappa(T', d)\, \frac{O^2}{2k'}. \tag{2}$$

At $\eta = 1$ this reduces to the single-skill $\min_j w_j/p_j$. The ungridded constant $w_j \eta / p_j$ would overestimate when the round-up engages, so the grid-aware form is correct.

Composition. Job sets of trade-disjoint groups are disjoint, so their bounds add. The implementation takes

$$\text{LB}(s) = \max\Big(\sum_{g \in G} \max\big(\text{LB}_{\text{i}}(A_g),\ \max_d \text{LB}_{\text{ii}}(\{g\}, d)\big),\ \max_d \text{LB}_{\text{ii}}(G, d) \Big), \tag{3}$$

where the global group term is evaluated only under skill overlap, so at L0 the bound reduces exactly to the single-skill per-trade scan. A contiguous chain-segment refinement (partitioning the cycle by a small dynamic program) is available but unexercised, because the per-trade and global-group terms did not exhibit the loose-bound plateau that would motivate the refinement.

Verification. Three released tests exercise the bound. First, the shaped return telescopes to $-\mathrm{TWT}/100$ at machine precision on random flexible episodes. Second, the bound equals the single-skill bound at L0 with $\eta = 1$ on 300 randomized states. Third, on 50 small static snapshots across the ladder the bound never exceeds the CP-SAT optimum. Tightness is not certified per cell. The admissible constant shrinks under heavy-tailed durations, and the per-job term weakens as eligible sets grow, so the bound is loosest in penalized full-flexibility states. Shaping exactness is unaffected, because the return telescopes to $-\mathrm{TWT}$ whatever the bound's tightness. In the penalized full-flexibility cells, the tested increase in training budget did not improve the policy pool (Robustness and Ablations section of the main manuscript). A state-resolved audit needs care, because released-order snapshots at a decision instant cover only part of the eventual workload. Bound-to-optimum ratios are therefore meaningful only where the remaining workload is fully released. On the static snapshots, which meet that condition, the bound at the start state recovers a mean 0.56–0.66 of the CP-SAT certified optimum across the ladder structures and both η values. Because the bound is admissible, this ratio is at most 1 by construction. At this level the bound is tight enough to shape the reward and loose enough that the shaping does not replace the policy's own optimization. The reward-shaping ablation and a comparison against a simpler per-trade bound are the two direct next checks.

S3 Secondary Metrics and Latency Accounting

This text reports the secondary metrics, the per-decision latency accounting, and the $\eta = 1.0$ companion to the dynamic main table of the Prediction Test section of the main manuscript. It closes with the full forensics of the uncertified static instances that the Static Reference section summarizes in two sentences.

Each released result row records makespan, mean flow time, overall SLA-breach share, per-priority breach shares, per-decision latency, and per-replan wall time. On the Tier-1 grid the ranked rules take 0.041 ms per dispatch decision (median), and the verdict-class policy takes 0.45 ms on CPU. The rolling solver spends 0.1 s per replan on its subsample. A deployment compares these times with its own response-time requirement. Makespan, mean flow time, and per-priority breach shares track weighted tardiness across the ladder without reordering the method comparison.

On the headline contended family ($m = 0.6$, $\eta = 0.8$) the winning rule misses its due dates on 1.30% of orders with dedicated crews, 1.26% on the chain, and 1.27% under full flexibility; the verdict-class policy pool follows the same shallow dip (1.41%, 1.30%, 1.39%). So SLA-breach share moves with weighted tardiness rather than against it, easing by the same small margin, lowest on the chain and edging back up under full flexibility.

Uncertified Static Instances: The Three-Way Split

The static reference of the main manuscript reports a bound-based gap over every instance whose solver lower bound is informative, that is, non-zero. That subset is every instance outside the FULL ($\eta = 0.8$) row, where the bound omits the hard instances it cannot bracket and only 64% of instances certify within 300 s. The 131 uncertified instances in that row split three ways. 48 end with zero tardiness and are therefore optimal anyway. A further 10 carry positive tardiness that the 300 s lower bound brackets to within a rounding error, so they are near-optimal and sit inside the informative-bound column of the static reference table. Only the remaining 73 have a bound that stays at zero and

Table S3. Dynamic table at $\eta = 1.0$ (Tier-1), completing the $\eta = 0.8$ block of the dynamic main table in the Prediction Test section of the main manuscript: pooled seed-mean TWT by structure and crew multiplier. At $\eta = 1.0$ the efficiency penalty is switched off, so several rules coincide. Policy rows are 10-seed means (std in parentheses); rules are deterministic. $n = 763$ instances per cell. Best method per column in bold; several rules coincide at $\eta = 1.0$ (see the Methods Under Test section of the main manuscript), and the bold marks their shared value.

	$m = 0.8$			$m = 0.6$		
Method	L0	CHAIN	FULL	L0	CHAIN	FULL
EDD	**318.45**	315.03	**314.37**	331.32	316.54	**314.40**
WSPT	321.70	317.07	315.05	335.54	321.80	316.18
ATC	318.95	315.59	314.37	328.99	318.29	314.40
pFIFO	318.45	315.03	314.37	331.32	316.54	314.40
LPT	336.35	316.75	314.66	414.28	331.13	315.86
LFJ-ATC	318.95	315.59	314.37	**328.97**	318.28	314.40
ATC-η	318.95	315.59	314.37	328.99	318.29	314.40
Policy (pair-MLP)	321.83 (2.02)	**314.66 (0.25)**	314.39 (0.01)	341.26 (4.93)	**316.30 (0.44)**	314.41 (0.02)
Other class	321.21 (1.29)	314.83 (0.58)	314.40 (0.03)	339.94 (4.95)	316.52 (1.34)	314.52 (0.26)
Random	326.74	317.32	315.35	359.28	324.28	316.85

brackets nothing; on those 73 the solver also returns no incumbent of its own within budget (the rule schedule enters only as a warm-start hint, which CP-SAT may discard), so the rules' distance from optimum there is genuinely unknown and the near-optimality reading does not cover them. Across all cells the 300 s re-run raises the lower bound on 74 of 297 of the instances not proved at 60 s and leaves it unchanged on the rest, so the certificate frontier, not the primal side, is what stalls.

S4 Locked Defaults: Hyperparameters, Seeds, and Budgets

This text lists every default that the Methods Under Test and Experimental Setup sections of the main manuscript describe only in summary: the solver, metaheuristic, and policy settings, the grids and seeds, the full pair-token feature inventory, and the hardware and software the experiments ran on.

All single-skill defaults carry over verbatim. SLA windows 8/24/80/171.4 bh for P1–P4 set the due-date offset per class; the weights 8/4/2/1 price a unit of tardiness. The p95-weekly-hours crew estimator sizes each trade's technicians, and the 1,000-order MISC threshold merges low-volume trades into one shared pool. The CP-SAT budgets (60 s, 300 s tail, two workers, warm starts as hints) bound the offline search. The rolling settings (2 s, two workers, 0.1 bh arrival batching, 4 bh staleness trigger, flow-time tie-break) set the recompute cadence. The GA settings (population 100, 60 s, OX, swap 0.2, tournament 3, elitism 2, stall 200) control the metaheuristic baseline's search. The PPO settings (lr 3×10^{-4}, $\gamma = 1.0$, GAE $\lambda = 0.98$, clip 0.2, 4 epochs, minibatch 1024, entropy 0.01, value coefficient 0.5, gradient clip 0.5, 16 environments $\times$ 512 steps) fix the policy-update rule.

The flexible regime adds settings, all locked in the dated protocol log before any verdict run. The η grids (main $\{1.0, 0.8\}$, sensitivity $\{0.9, 0.75\}$; $\eta = 1$ is exactly penalty-free) scale the efficiency penalty on secondary-skill work. The φ grid $\{0.25, 0.5, 1.0\}$ sets the cross-training reach along the chain. The workload-descending chain-order rule, the overlay construction seed 20260707, and the chain-permutation seeds 20260708–20260710 fix the chain links and the overlay draw. The technician tie-break (primary, least-flexible, lowest id) and the LFJ-ATC tie tolerance 1% decide equal-score assignments. The candidate cap of 64 orders per trade (ablation 256), the signature-deduplicated technicians, and the 256-slot pair tensor with deterministic slack-first truncation together bound the order-technician pairs scored each step. Training runs 1,200 policy updates; averaged seeds come from pools pair-MLP 301–310, pair-attention 401–410, specialists 501–510 (chain) and 601–610 (full). The uniform mixed-flexibility curriculum

materializes instances at $m = 1$ with the overlay carrying m. The Holm correction for the Gate P family controls the family-wise error rate. The gate thresholds are those stated verbatim in the Experimental Setup section of the main manuscript.

Pair-Token Feature Inventory

Each pair token is a 38-dimensional vector: the 12 single-skill order features (log processing time, slack, unavoidably-late flag, class weight, class one-hot, preventive flag, waiting time, share of trade's queued work, WSPT index); 6 pair features (primary-skill flag, log realized pair duration, technician flexibility $|S_u|$, share of the technician's skills with non-empty queues, the technician's primary-trade queue pressure, and a last-idle-eligible-technician flag); and 20 context features (queue length and queued work per technician for the three busiest reachable trades, smallest reachable slack, share of reachable queue in the top two priority classes, share of technicians idle, a corrective-arrival rate EMA (exponential moving average), week-position sine and cosine, episode progress, and flexibility descriptors: structure one-hot, φ, η, and budget per head B/N). Normalization constants are fixed, as in the single-skill benchmark.

Hardware and Software

All experiments ran on one 24-core workstation (64 GB RAM) with a single NVIDIA RTX PRO 5000 GPU used for policy optimization. Evaluation and solving are CPU-only (evaluation used up to 24 concurrent worker processes; the GPU was shared with unrelated workloads). Policy optimization across all seeds and classes consumed roughly 161 GPU-hours in total, reported as a contended wall-clock upper bound since the GPU was shared with unrelated workloads. Each CPU-only dispatching decision is priced in well under a millisecond (Text S3). The stack is Python with OR-Tools CP-SAT and PyTorch. Exact versions, seeds, and scripts ship with the repository, and a single script rebuilds every figure and table from the released result files.

Training Diagnostics

Fig. S1 carries the development curves and the checkpoint selection for both architecture classes, summarized in the Robustness and Ablations section of the main manuscript.

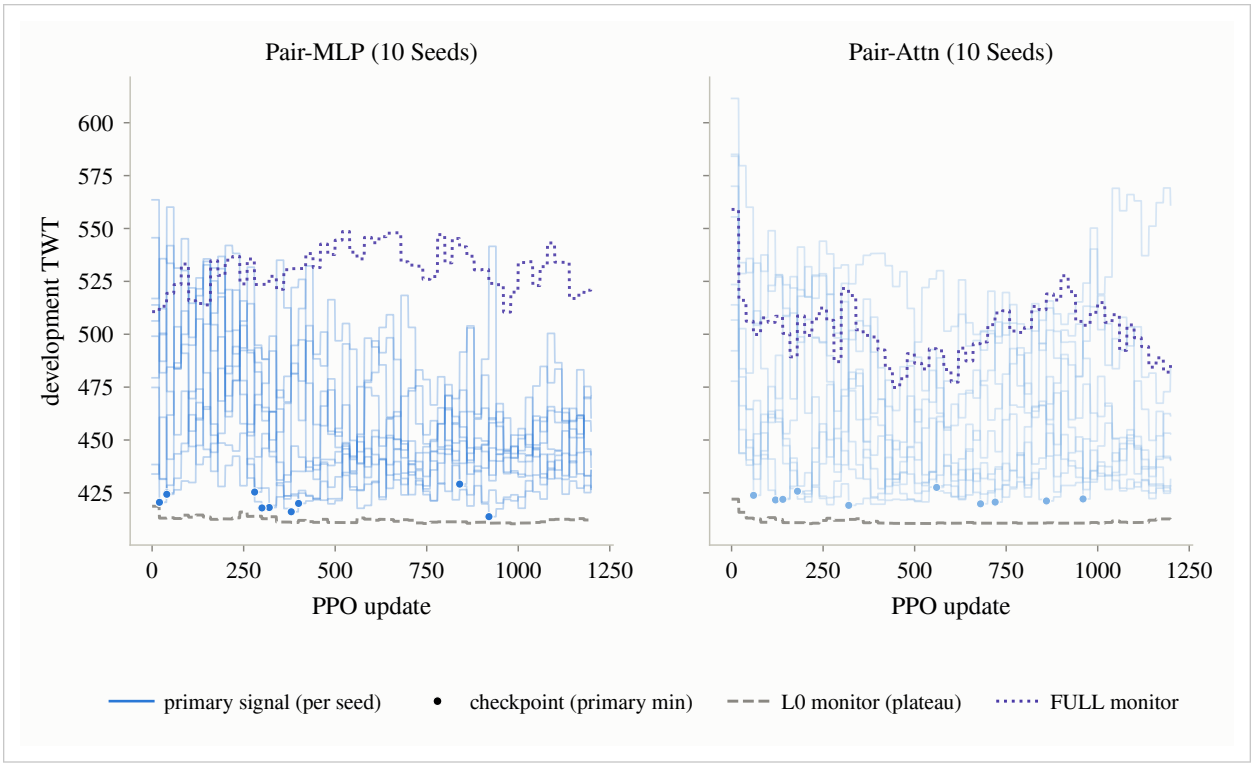


Figure S1. Training diagnostics: primary development signal (tight-capacity chain cell) and monitors per update, 10 seeds per class, checkpoint at the per-seed primary minimum. Takeaway: the tight-capacity development signal discriminates checkpoints that the default-capacity monitors cannot.

S5 Statistical Detail

This text gives the statistical detail behind the tests summarized in the Experimental Setup section of the main manuscript: the released per-family p-value tables, the resampling schemes and the dependence structure they answer to, the treatment of the method-selection step in envelope intervals, and the scope the pooled p-values may be read at. It closes with the rank-correlation robustness figure.

The released analysis files include, per Gate P scope, the seven paired Wilcoxon statistics with raw and Holm-adjusted p-values; per-seed pooled means against the best rule; and win/tie/loss tallies at the $\varepsilon = 1.0$ tie tolerance for every method pair in every family. Direction matters as much as significance here: the policy's Holm-significant wins are limited to LPT (m=0.6); WSPT (pooled, m=0.6); every other significant comparison reads in the rule's favor.

Resampling Schemes and the Dependence Structure

The elementary observation is one method on one instance-configuration. Configurations nest in base instances (one replayed window under several overlays), which in turn nest in campuses. Pooled scopes weight every instance equally, so campuses contribute in proportion to their instance counts (192, 171, 200, and 200; 400 instances of 150 orders and 363 of 400). Within a size class every test window has a distinct calendar start. Across size classes, 296 windows share a start, because the 400-order window extends the 150-order one. The base-instance key includes the size class, so the base-instance bootstrap treats these shared-start windows as independent. Because that is the less conservative choice, the headline capture ratio is also re-estimated with a stricter block bootstrap that resamples (campus, calendar-start) groups, binding the two sizes into 467 blocks instead of 763 instances. The interval widens only slightly, to 0.874

Table S4. Gate P paired Wilcoxon tests: policy seed-mean vs each ranked rule, raw and Holm-adjusted p-values per scope, and the winner of each comparison on the pooled means (R = rule lower, P = policy lower). Paired effect sizes (rank-biserial, with CIs) for the pooled scope are in the Prediction Test section of the main manuscript and the released tables. n = common instance-configurations per scope.

	pooled			m = 0.8			m = 0.6		
Rule	raw	Holm	win	raw	Holm	win	raw	Holm	win
EDD	<0.001	<0.001	R	<0.001	<0.001	R	<0.001	<0.001	R
WSPT	0.005	0.005	P	<0.001	<0.001	R	<0.001	<0.001	P
ATC	<0.001	<0.001	R	<0.001	<0.001	R	<0.001	<0.001	R
pFIFO	<0.001	<0.001	R	<0.001	<0.001	R	<0.001	<0.001	R
LPT	<0.001	<0.001	R	<0.001	<0.001	R	<0.001	<0.001	P
LFJ-ATC	<0.001	<0.001	R	<0.001	<0.001	R	<0.001	<0.001	R
ATC-η	<0.001	<0.001	R	<0.001	<0.001	R	<0.001	<0.001	R

n = pooled: 6104 / m08: 3052 / m06: 3052.

(95% CI 0.763–0.934), and stays entirely above the 0.70 threshold, so the shared-start dependence does not carry the verdict.

The per-configuration observations are not independent, since they share base order streams, so post-hoc every headline estimate is also reported with a 95% cluster-bootstrap interval that resamples base instances within campus (10,000 resamples), and with a coarser whole-campus variant alongside. The whole-campus variant is a descriptive range rather than a calibrated interval, because four campuses admit only 35 distinct campus resamples. Paired effect sizes (rank-biserial correlation) accompany the Wilcoxon p-values.

Method Selection Inside the Bootstrap

An envelope estimate takes, for each structure, the lowest pooled mean TWT among the tested methods, so its bootstrap must either repeat this selection or hold it fixed. The released intervals repeat it. Each resample recomputes every method's pooled mean and takes the lowest per structure, so the interval includes the uncertainty about which method is selected. The alternative conditions on the original sample, in which the envelope selects LFJ-ATC at L0, the pair-MLP policy pool for CHAIN(1.0), and EDD for FULL (m = 0.6, η = 1.0). That conditional interval resamples only the values of these three methods. The original selections recur in only 55%, 69%, and 58% of the 10,000 resamples, respectively, because the leading methods' pooled means lie close together relative to the resampling variation. The headline capture ratio is nevertheless almost insensitive to this choice. It reads 0.870 (95% CI 0.794–0.932) with re-selection and 0.870 (95% CI 0.784–0.932) conditional on the original selections, so the two intervals differ only at the lower end and both lie above the 0.70 threshold. Fixed-EDD estimates involve no selection step and have a single interval. Every envelope interval in Tables S15, S16, S20, and S22 re-selects in the same way. The bars of the design-curve figure of the main manuscript (Fig. 2) hold each structure's selected method fixed, so they show the conditional form.

What the Pooled p-Values Do and Do Not Support

The large pooled-scope n (for example 6,104 instance-configurations in the pooled Gate P scope) counts configurations, not independent organizations: the same base workload recurs under several overlays, crew multipliers, and efficiencies, and only four verdict campuses enter. The pooled p-values are therefore descriptive summaries over *the work windows in this released corpus*, not population-level inference over facility portfolios; the cluster and calendar-start bootstraps, the leave-one-campus-out re-derivations, and the per-campus and per-size splits reported in the Results section of

the main manuscript and in Text S6 are the primary uncertainty statements. Campus-level generalization is bounded accordingly, as the Discussion section of the main manuscript states.

Robustness of the Method Ranking

Fig. S2 carries the fifteen sensitivity variants behind the rank-correlation summary of the Robustness and Ablations section of the main manuscript.

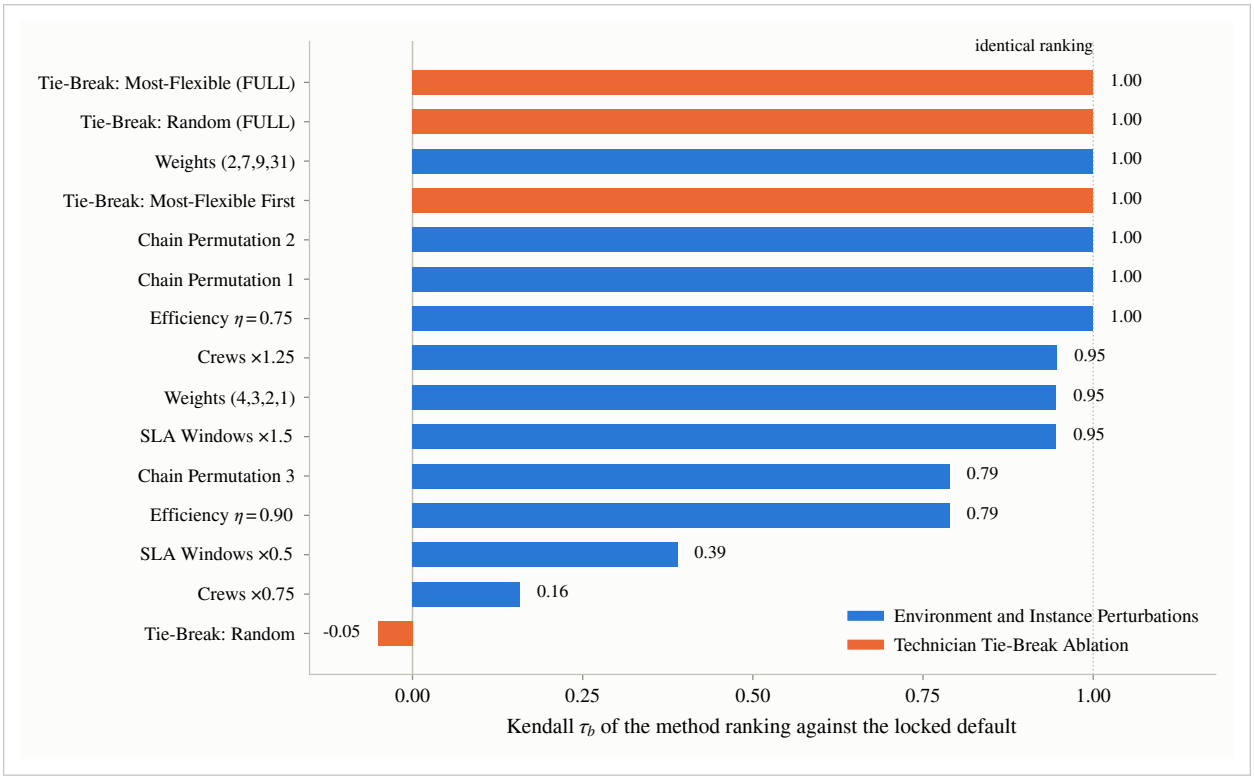


Figure S2. Robustness of the method ranking: Kendall τ_b between the ranking under each sensitivity variant and the locked default, designated cells, n = 120 per variant, shown as ranked bars. Takeaway: rankings survive the efficiency, weight-vector and chain-order perturbations, move only where the perturbation forces deeper overload, and collapse when the technician tie-break is randomized.

The chain result is also stress-tested on the full verdict set, because rank correlations can stay high while effect sizes drift. The chain order is not what carries it: across ten seeded chain permutations the rules-envelope capture spans 0.842 to 0.894 (median 0.856), and a chain built on coarse trade-skill adjacency captures 0.881 (fixed-EDD 0.897) against the released chain's 0.853, so the chain may follow recognizable skill families at no cost. Which half of each trade receives the second skill does not change the outcome at all. Across the lowest-indexed assignment and three independent random technician selections, the pooled weighted tardiness is identical to four decimal places, because within a trade every cross-skilled technician carries the same successor skill and is interchangeable: the choice moves technician identities but not the set of jobs each skill can cover. The half-adoption dividend is therefore not merely stable across selections, it is exactly invariant to them. Nor is the scalar penalty load-bearing: seeded per-trade-pair draws from $[0.70, 0.95]$ in place of $\eta = 0.8$ (realized means 0.806–0.832) leave the full-flexibility dividend at or above its uniform value in all three draws and move the chain dividend to between 3.33 and 5.35 weighted units against the uniform 4.11, and the chain-matches-full reading survives 3/3 draws under fixed EDD and 2/3 under the envelope, the

one exception being a 0.09%-of-L0 tie.

S6 Data Contract, Reuse Boundary, and Descriptives

This text states what a facility portfolio must be able to export before the framework can run on it, what this paper inherits from the single-skill corpus release and what it adds, and the workload, campus, and cost descriptives behind the Framework and Design Curve sections of the main manuscript.

Minimum data contract. Table S5 lists every field the framework consumes. The preprocessing that produces them from a raw CMMS export ships with the corpus release (Zhang and Ku Chia 2026): duplicate and canceled-order handling, duration validation, the priority-class mapping, and the concatenated business-hour axis. This paper adds no preprocessing of its own, so any portfolio whose CMMS can export these fields can run the full sweep. Where a real roster export exists, it replaces the p95 calibration and upgrades the [C] layer to observation.

Table S5. Minimum CMMS data contract of the framework: six work-order fields, one crew table, and the designed inputs an analyst declares. Provenance tags as in the Framework section of the main manuscript: [R] replayed fact, [C] calibrated from the log, [D] designed input. $n = 10$ fields.

Input	Field	Tag	Used for
Work order	identifier	[R]	tracking, tie-breaks
	trade / craft code	[R]	eligibility, crews
	release timestamp	[R]	arrival stream
	labor duration	[R]	processing time p_j
	priority class	[R]	weight w_j
	due date or SLA class	[R]	tardiness
Crew table	p95 weekly hours per trade	[C]	crew sizes, chain order
Designed	structures Λ, φ, m, η	[D]	the design sweep
	trade adjacency / licensing	[D]	feasible wirings
	cost accounting	[D]	frontiers (Table S9)

Reuse boundary against the single-skill benchmark. The framework deliberately holds the single-skill benchmark (Zhang and Ku Chia 2026) fixed wherever reuse strengthens comparability, and Table S6 states the boundary explicitly. No experimental result of this paper appears in that release or elsewhere. The shared material is the input corpus and the protocol conventions, and every number reported here comes from the multi-skill runs of this paper. The manuscript is self-contained: the model, the overlay constructions, the method suite, and the gates are fully specified in the Problem Formulation through Experimental Setup sections of the main manuscript together with these supplemental texts. The citation to the corpus is needed only to obtain the input data and to verify the regression anchor.

Table S6. Reuse boundary: what is inherited unchanged from the single-skill corpus release, what this paper extends, and what is new. $n = 3$ columns of components.

Reused unchanged	Extended here	New here
Work-order instances (byte-identical); crew calibration; business-hour axis; priority weights; locked PPO defaults and seeds; statistical conventions (paired Wilcoxon, tie tolerance)	Validator (skill legality, pair durations); dispatch engine (order selection to pair selection); six priority rules; CP-SAT model (signature symmetry breaking); GA decoder; rolling solver snapshots; Holm correction added	Resource-overlay layer and ladder; topology controls (pairs, random, hub, eligibility-constrained); two flexibility-aware rules; patient rules; pair-scoring policies and features; admissible bound for overlapping skill pools; Gates C and P; regression anchor; both frames; every experiment and number

Workload descriptives and campus-level detail. The three generated tables below carry the per-campus descriptives behind the crew-multiplier mapping of the Framework section of the main manuscript, the campus-level robustness detail behind the Design Curve section, and the cost frontiers of that same section. The per-trade descriptive table (crew, p95 hours, and utilization at each multiplier for every trade) ships with the repository.

Table S7. Workload and roster descriptives per verdict campus. Crews are calibrated on p95 weekly labor hours [C]; the table maps each crew multiplier m to its operational regime: campus-wide p95-week utilization (demand of a 95th-percentile week over scaled roster hours), the median and maximum per-trade ratio, and the share of trades whose p95-week demand exceeds their scaled crew. $n = 4$ campuses; the per-trade table ships with the repository.

Campus	K	N	m	p95-week util.	trade median	trade max	trades over
5	11	116	1.0	0.95	0.87	1.00	0%
			0.8	1.19	1.09	1.32	73%
			0.6	1.60	1.57	1.99	100%
9	13	99	1.0	0.92	0.87	0.99	0%
			0.8	1.15	1.08	1.23	77%
			0.6	1.51	1.44	1.70	85%
10	13	154	1.0	0.95	0.91	0.99	0%
			0.8	1.19	1.22	1.25	77%
			0.6	1.61	1.57	1.82	100%
12	13	186	1.0	0.97	0.97	1.00	0%
			0.8	1.23	1.23	1.31	100%
			0.6	1.61	1.57	1.77	100%

Replay test windows release work at a median intensity of 0.35 (150-order) and 0.44 (400-order instances) of full-roster capacity (90th percentile 0.59 and 0.83); 172 short windows release near-instantaneously. Contention in the replayed weeks is episodic, which is exactly the regime the p95 calibration prices.

Table S8. Campus-level detail for the evaluable family ($m = 0.6$, $\eta = 1.0$): capture ratio and full-flexibility dividend per single campus, with each campus left out, and per instance size. Envelope = the rules envelope (seven ranked rules and Random), as in the topology and feasibility panels; EDD = fixed dispatcher. n per campus: 192, 171, 200, 200 instances.

Scope	ρ (env.)	ρ (EDD)	Δ(FULL) (env., units)
campus 5 alone	0.99	1.00	4.9
campus 9 alone	0.99	0.99	8.2
campus 10 alone	0.94	0.95	9.4
campus 12 alone	0.77	0.77	33.1
without campus 5	0.84	0.84	17.8
without campus 9	0.83	0.86	16.2
without campus 10	0.84	0.86	16.4
without campus 12	0.97	0.98	7.7
150-order instances	0.89	0.90	5.5
400-order instances	0.84	0.87	24.5

Capture-Ratio Detail Behind the Campus Table

The Design Curve section of the main manuscript summarizes five checks establishing that the capture result is not an artifact of how it is computed, and points here for the detail of four of them.

First, the cluster bootstrap over base instances puts ρ at 0.870 (95% CI 0.794–0.932), with the coarser whole-campus resampling alongside at 0.870 (range 0.854–0.989). Four campuses admit only 35 distinct campus resamples, so that variant is a descriptive range rather than a calibrated interval. The same instance bootstrap puts the implementable fixed-EDD ratio at 0.874 (95% CI 0.781–0.927). Every interval sits entirely above the 0.70 threshold, so the chaining-suffices verdict would survive any capture threshold up to the interval floors.

Second, no single campus carries the result. Leaving any one campus out keeps the rules-envelope ratio (the

Table S9. Service against cost under three cost accountings (pooled verdict campuses at $m = 0.6$; TWT on the rules envelope). B counts added skill memberships and E enrolled technicians; scenario costs are relative units: uniform = B; enrollment = $E + B$ (a fixed per-technician enrollment cost plus a marginal per-membership cost); family = 1 per same-cluster membership, 2 per cross-cluster one. $n = 763$ instances per TWT. No structure wins every accounting: the generalist pool is cheapest under enrollment costs and, giving $K - 1$ skills to a few technicians, also holds its penalized dividend (TWT 324.2 at $\eta = 0.8$, marginally below the chain's 324.9); the eligibility-constrained chain is cheapest under family costs; and among the one-secondary-skill-per-technician wirings the closed chain retains the largest penalized dividend (Table S12).

Structure	B	E	uniform	enrollment	family	TWT η=1	TWT η=0.8
L0	0	0	0	0	0	329.0	329.0
CHAIN(1.0)	332	332	332	664	563	316.5	324.9
PAIRS	332	332	332	664	514	316.1	327.6
FEAS	332	332	332	664	463	319.6	326.6
GEN	334	29	334	363	607	316.1	324.2
FULL	3846	332	3846	4178	6978	314.4	325.8

eight-method estimator, the seven ranked rules and Random, that reads 0.853 pooled) within 0.83–0.97 (fixed-EDD 0.84–0.98), and each campus alone gives 0.77–0.99 (Table S8).

Third, weighting the four campuses equally rather than by instance count gives a macro-averaged rules-envelope capture of 0.92 (fixed-EDD 0.93). Both clear the 0.70 gate, so the chaining-suffices verdict survives equal campus weighting.

Fourth, both instance sizes agree. The 150-order and 400-order instances give the same reading separately (Table S8), which matters because 296 of the 763 test windows share a calendar start with the other size class and are not independent draws across sizes.

S7 Main-Text Exhibits in Full

This text carries exhibits summarized in the main manuscript: the experiment-family grid of the Experimental Setup section (Table S10), the static-reference table (Table S11), the same-budget topology panel (Table S12), the eligibility-scenario panel (Table S13), the held-out transfer table (Table S14), the flexibility-ladder illustration (Fig. S3), the flexibility-gradient figure of the Prediction Test section (Fig. S4), the decision map (Fig. S5), and the stress-test rescue on the held-out campus 2 (Fig. S6).

Table S10. Experiment families (verdict campuses unless stated; n per cell as listed; C(φ) abbreviates CHAIN(φ) in table cells throughout). Full grids, seeds, and designated subcells are recorded in the protocol log shipped with the repository.

Family	Track	Grid	Purpose
E1′ static	replay+gen, 15/(campus,size,track)	$\Lambda \in$ {L0, C(1.0), FULL}, $\eta \in \{1, .8\}$, m=1	RQ4 optima, gaps, hardness
E2′ Tier-1	replay, all test inst.	$\Lambda \in$ {L0, C(1.0), FULL}, $\eta \in \{1, .8\}$, $m \in \{1, .8, .6\}$	Gate P cells; decision map
E2′ Tier-2	replay, all test inst.	$\Lambda \in$ {C(.25), C(.5), GEN}, $\eta \in \{1, .8\}$, $m \in \{1, .6\}$	φ-curve; chain vs GEN
E3′ Frame U	generator, 30/(campus,U)	$U \in \{.9, 1, 1.1, 1.3\}$, Tier-1 $\Lambda \times \eta$, m=1	matched aggregate load
E4′ transfer and stress test	replay, campuses 1, 2	Tier-1 grid	RQ5 campus-2 rescue
E5′ sensitivity	designated subcells	see protocol log	Kendall τ_b robustness

Table S11. Static reference (E1′): CP-SAT proof rates, mean optimality gaps of the best ranked rule, and solve walls per structure at $m = 1.0$. Rule and GA gaps are computed on the certified subset of each cell, against the certified optimum; the last column is the best rule's mean gap to the strongest solver lower bound over every instance with an *informative* (non-zero) bound, an upper bound on its true optimality gap. Vacuous bounds (zero against positive tardiness) are excluded: none outside the FULL (0.8) row and 73 there, so that row's figure speaks to the informative-bound subset only. $n = 360$ instances per (structure, η) cell (15 per stratum over the verdict campuses × sizes 50/150/400 × replay/generator tracks).

Λ (η)	proved 60 s	proved 300 s	median wall (s)	best-rule gap	GA gap	rule vs bound (inform.)
L0	100%	100%	0.8	0.06%	0.00%	0.06%
CHAIN(1.0) (1.0)	100%	100%	1.7	0.01%	0.00%	0.01%
CHAIN(1.0) (0.8)	83%	100%	8.3	0.72%	0.00%	0.17%
FULL (1.0)	99%	100%	8.5	0.01%	0.00%	0.00%
FULL (0.8)	36%	64%	56.3	0.00%	0.00%	0.00%

Table S12. Same-budget topology panel at $m = 0.6$: every row grants one secondary skill per technician (B = headcount, identical to CHAIN(1.0)); only the trade-level wiring varies. Components = weakly connected components of the trade digraph (range over campuses); coverage = share of p95 workload in trades that receive at least one helper trade; ρ = capture ratio on the rules envelope and under fixed EDD at $\eta = 1.0$; the last column is the absolute dividend of the structure at $\eta = 0.8$ in weighted units (the $\eta = 0.8$ denominators are too small for ratios). GEN concentrates the same membership count in full generalists; OPT is the hill-climbing best-found map, searched on the training windows only and evaluated once here, its coverage being the searcher's own choice. $n =$ 763 instances per cell. Takeaway: at full secondary speed, capture tracks workload coverage and not connectedness (PAIRS matches the chain with five to six disconnected components); under the 20% penalty the closed chain is the one-secondary wiring that loses least; and searching for a bespoke wiring on the portfolio's own history buys nothing out of sample over the simple full-coverage chain.

Structure	Comp.	Coverage	ρ (env.)	ρ (EDD)	Δ at η=0.8
CHAIN(1.0)	1	1.00	0.85	0.87	4.1
Adjacency chain	1	1.00	0.88	0.90	4.3
GEN	1	1.00	0.88	0.90	4.7
PAIRS	5–6	1.00	0.88	0.90	1.4
RAND1 (3 draws)	1	0.36–0.79	0.19–0.70	0.30–0.64	–
FEAS (eligibility-constrained)	1	0.61	0.64	0.67	2.4
STAR (hub)	1	0.51	0.18	0.13	−7.6
OPT (best-found, train)	1	0.54–0.77	0.79	0.82	−1.5

Table S13. Illustrative feasibility scenarios for the one-secondary-skill chain at $m = 0.6$: how the capture ratio moves as an eligibility constraint tightens. These scenarios are designed inputs, not measured regulation; a facility would substitute its own eligibility matrix. Coverage = share of p95 workload in trades that may receive a helper; capture is on the rules envelope at $\eta = 1.0$; Gate C requires $\rho \geq 0.70$. The chain suffices under the unrestricted and moderately constrained scenarios and tracks coverage down as the restrictive scenario removes it. $n = 763$ instances per cell; verdict campuses.

Illustrative scenario	Example ineligible trades	Coverage	ρ (env.)	Gate C
Illustrative unrestricted	every trade cross-trainable	100%	0.88	passes
Illustrative moderately constrained	e.g. electrical, elevators	88%	0.87	passes
Illustrative restrictive	e.g. + plumbing, fire	61%	0.64	fails

Table S14. Transfer to the held-out campuses (E4′): TWT of the best observed method and its ratio to the dedicated baseline, by structure, at $\eta = 0.8$; campus 1 is well provisioned, campus 2 chronically under-provisioned. Policy columns are 10-seed means. n per cell in the released tables.

	$m = 0.8$		$m = 0.6$	
Campus, Λ	TWT	ratio	TWT	ratio
1, L0	140.93	1.00	143.59	1.00
1, CHAIN	143.53	1.02	147.29	1.03
1, FULL	149.53	1.06	159.46	1.11
2, L0	2879.66	1.00	3561.25	1.00
2, CHAIN	2197.66	0.76	2587.65	0.73
2, FULL	2072.19	0.72	2429.79	0.68

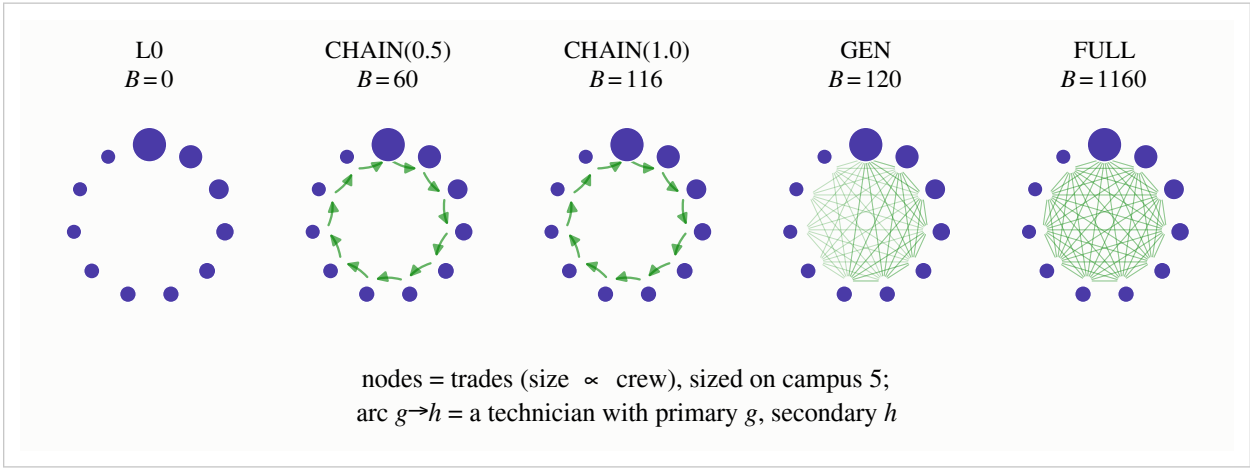


Figure S3. The flexibility ladder on one real campus, as trade-level eligibility structure: nodes are trades on a circle in chain order (size proportional to crew), and a directed arc $g \to h$ marks a technician with primary trade g and secondary skill h. L0 has no arcs; CHAIN(1.0) is a single directed cycle covering all trades; GEN is a hub of generalists; FULL is complete (the dense GEN and FULL meshes are drawn as headless chords for legibility). The realized skill-membership budget B is annotated; headcount is identical in every panel. Takeaway: the ladder spans dedicated to fully flexible at fixed headcount, with the complete 2-chain a single cycle that grants one secondary skill per technician.

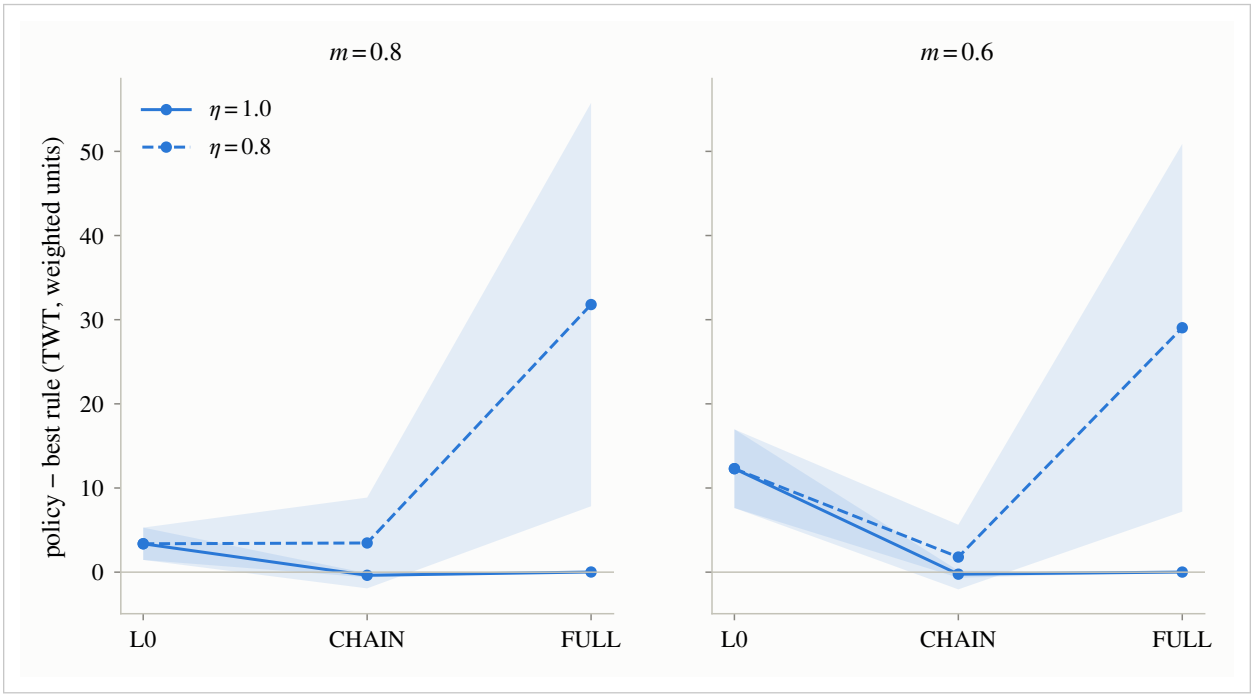


Figure S4. The flexibility gradient: policy-versus-best-ranked-rule TWT gap (seed mean, per-seed ribbon) as a function of structure, panels = crew multiplier, both η values. Positive gap means the policy is worse. The ribbon shows training-seed variability only; workload uncertainty is carried separately by the per-cell bootstrap intervals in the released tables. Policy numbers are 10-seed means; n per cell in the released tables. Gate P outcome: did not pass (branch B). Takeaway: the policy-versus-rule gap narrows on the chain and reverses in the chained cells at $\eta = 1$, but re-opens under full flexibility once the efficiency penalty bites.

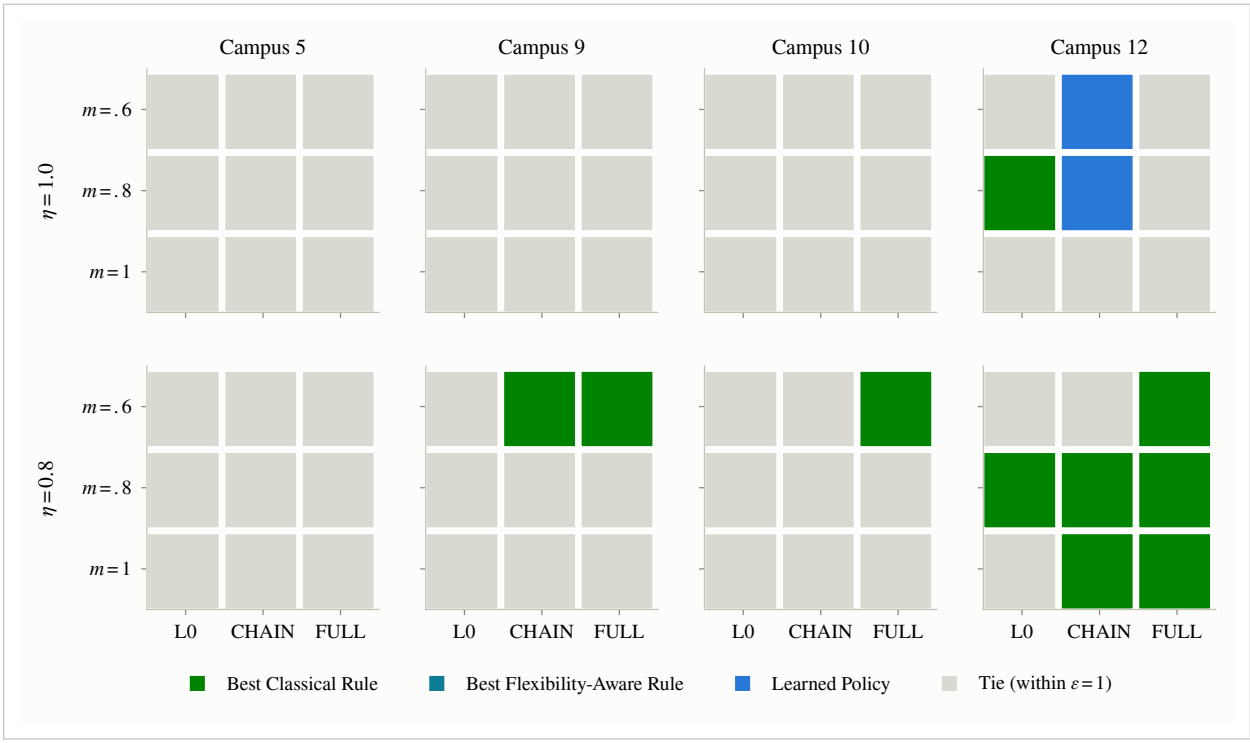


Figure S5. Decision map of the replay track. Each cell (crew multiplier × structure) shows its unique winner among the best classical rule, the best flexibility-aware rule, and the policy (10-seed mean), and it is a tie when two or more of these lie within $\varepsilon = 1$ weighted unit of the best. Panels are verdict campuses, and rows are η values. Takeaway: 33 of the 36 cells at $\eta = 1.0$ and 27 at $\eta = 0.8$ are ties. A classical rule wins uniquely in 1 cell at $\eta = 1.0$ and in 9 at $\eta = 0.8$, the learned policy in 2 chained cells at $\eta = 1.0$, and a flexibility-aware rule in none.

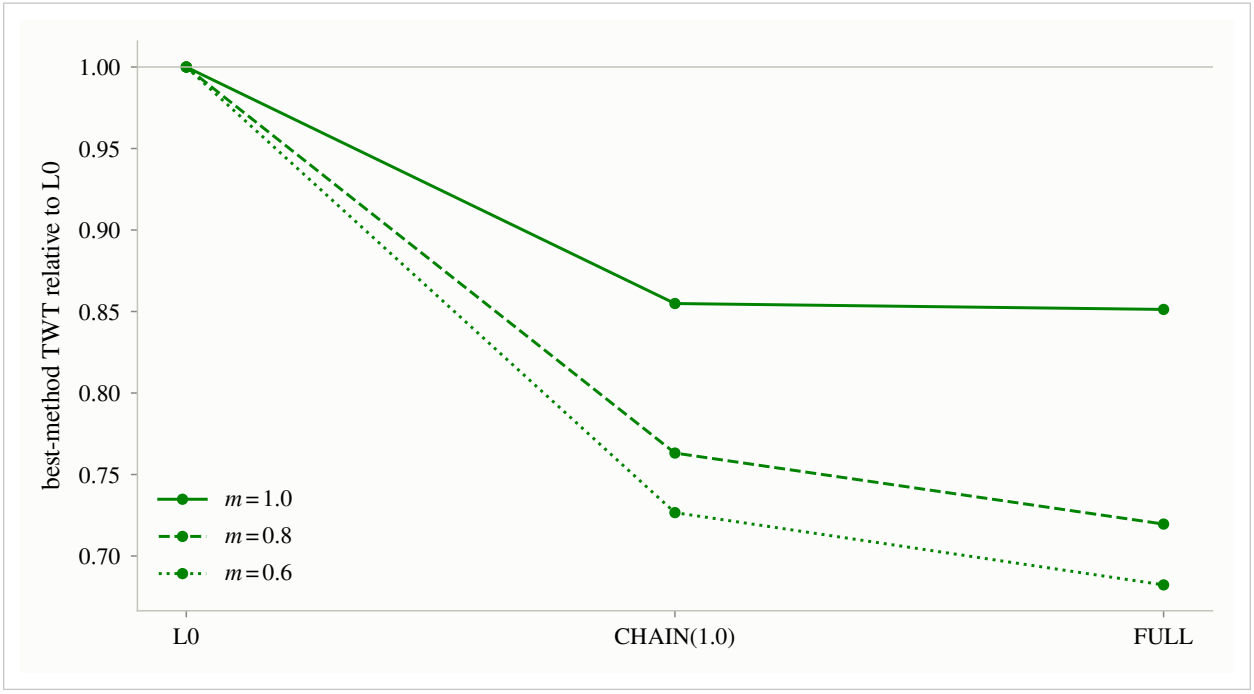


Figure S6. Stress test on the deliberately under-provisioned held-out campus 2: TWT of the best observed method relative to the dedicated baseline across structures, $\eta = 0.8$, three crew multipliers; n per cell in the released tables. Intervals are instance-level only, because the result comes from one campus. Takeaway: on this stress-test campus the simulation values cross-training at up to 32% lower weighted tardiness (complete chain 27%), so flexibility hedges a crew mis-calibration where it hurts most.

S8 Sensitivity Analyses, Hypothetical Crew Sizes, and Matched Aggregate Load

This text carries the analyses that re-read the pre-specified verdicts under changed assumptions. They vary the roster-calibration rule (Table S15) and restrict the order stream to its corrective-only and short-order subsets (Table S16). Other analyses draw the secondary-skill efficiency per technician (Table S17), move the gate thresholds (Table S18), and resolve the exact reference by instance size (Table S19). Two further analyses concern the order model and the second experimental frame. The crew-size analysis reads each long order as the work of a hypothetical crew (Table S20), and the Frame U tables compare the structures at matched aggregate offered load (Tables S21 and S22). Except for the pre-specified Frame U fallback, none of these analyses enters a gate. The Results and Discussion sections of the main manuscript state what each one does and does not change.

Table S15. Roster-calibration sensitivity. Crews are rebuilt from the raw corpus at four weekly-hours quantiles on the training window and at the 95th percentile on all recorded years, and the flexibility ladder is re-scored with the seven ranked rules and Random. N = scaled headcount and utilization = demand of a 95th-percentile week over scaled roster hours, both as ranges over the four verdict campuses. Δ(FULL) is the full-flexibility dividend on the rules envelope at $\eta = 1.0$, with its share of L0 in parentheses. A dash in a ρ column marks a family below the 2% evaluability threshold, so the ratio is not evaluable; where a ratio is shown, the parenthesis is a 95% instance-cluster bootstrap interval. $n = 763$ instances per cell.

Calibration	m	N	Util.	L0 TWT	Δ(FULL)	ρ (env.)	ρ (EDD)	Δ at η=0.8 (ch./full)
p_{90}, train	1.0	70–166	1.09–1.58	320.1	5.7 (1.8%)	–	0.95	2.5 / 2.0
	0.8	57–133	1.35–1.94	324.6	10.3 (3.2%)	0.90 (0.84–0.95)	0.92	4.0 / 3.0
	0.6	42–100	1.81–2.64	338.6	24.1 (7.1%)	0.86 (0.81–0.90)	0.89	7.4 / 2.9
p_{95}, train (released)	1.0	99–186	0.92–0.97	315.5	1.2 (0.4%)	–	–	−0.6 / −0.7
	0.8	79–147	1.15–1.23	318.5	4.1 (1.3%)	–	–	0.1 / −0.1
	0.6	60–112	1.51–1.61	329.0	14.6 (4.4%)	0.85 (0.78–0.91)	0.87	4.1 / 3.2
p_{95}, train (rebuilt)	1.0	99–186	0.92–0.97	315.5	1.2 (0.4%)	–	–	−0.7 / −0.7
	0.8	79–147	1.15–1.23	318.5	4.1 (1.3%)	–	–	0.0 / −0.1
	0.6	60–112	1.51–1.61	329.0	14.6 (4.4%)	0.85 (0.78–0.90)	0.87	4.0 / 3.2
$p_{97.5}$, train	1.0	119–207	0.63–0.87	315.0	0.7 (0.2%)	–	–	−0.8 / −0.5
	0.8	96–166	0.78–1.09	317.0	2.6 (0.8%)	–	–	−1.0 / 0.2
	0.6	72–125	1.03–1.45	319.8	5.4 (1.7%)	–	–	−2.2 / −2.4
p_{99}, train	1.0	138–342	0.32–0.79	314.9	0.5 (0.2%)	–	–	−0.5 / −0.2
	0.8	110–274	0.40–0.99	316.5	2.2 (0.7%)	–	–	−0.4 / 0.2
	0.6	83–206	0.54–1.32	318.8	4.4 (1.4%)	–	–	−3.0 / −0.5
p_{95}, all years	1.0	96–242	0.61–0.98	315.1	0.7 (0.2%)	–	–	−0.7 / −0.8
	0.8	77–194	0.76–1.21	316.0	1.7 (0.5%)	–	–	−1.4 / −1.6
	0.6	57–146	1.01–1.64	321.9	7.5 (2.3%)	0.74 (0.65–0.84)	0.74	−3.4 / −4.3

Table S16. Corrective work and short orders. Panel (a) attributes the weighted tardiness of the released cells at $m = 0.6$, $\eta = 1.0$ to each order class: shares of the 205,200 orders and of their labor hours, then of the L0 total and of each dividend, under the cell's envelope winner and under fixed EDD. Panel (b) drops one class from every instance, recalibrates the crews on the stream that remains and re-scores the ladder, so each reduced stream carries its own contention regime and its L0 level is not comparable across rows. A dash in a ρ column marks a family below the 2% evaluability threshold, so the ratio is not evaluable; a shown ratio carries a 95% instance-cluster bootstrap interval. $n = 763$ instances per cell (762 on the corrective-only stream, whose one all-preventive instance is empty).

(a) Where the tardiness and the dividend sit

			rules envelope			fixed EDD		
Order class	Orders	Hours	L0	Δ(chain)	Δ(full)	L0	Δ(chain)	Δ(full)
Corrective	52.5%	58.6%	99.7%	92.1%	93.2%	99.3%	84.0%	86.0%
Preventive	47.5%	41.4%	0.3%	7.9%	6.8%	0.7%	16.0%	14.0%
$p_j \leq 8$ bh	90.7%	43.0%	1.3%	31.0%	29.3%	2.7%	58.5%	53.5%
$p_j > 8$ bh	9.3%	57.0%	98.7%	69.0%	70.7%	97.3%	41.5%	46.5%

(b) Each stream with crews recalibrated on it

Stream	m	Orders	L0	Δ(FULL)	ρ (env.)	ρ (EDD)	Δ at η=0.8 (ch./full)	
Full stream (released)	1.0	205,200	315.5	1.2 (0.4%)	–	–	−0.6	−0.7
	0.8		318.5	4.1 (1.3%)	–	–	0.1	−0.1
	0.6		329.0	14.6 (4.4%)	0.85 (0.78–0.91)	0.87	4.1	3.2
Corrective only	1.0	107,668	322.5	7.7 (2.4%)	0.92 (0.86–0.98)	0.89	3.4	3.1
	0.8		330.3	15.5 (4.7%)	0.89 (0.85–0.96)	0.90	7.7	7.9
	0.6		371.3	56.4 (15.2%)	0.86 (0.83–0.89)	0.90	29.5	38.3
$p_j \leq 8$ bh only	1.0	186,119	4.3	4.3 (99.5%)	0.55 (0.38–0.75)	0.59	1.6	3.0
	0.8		6.8	6.8 (99.2%)	0.46 (0.32–0.67)	0.52	1.8	4.5
	0.6		18.8	18.5 (98.6%)	0.42 (0.36–0.55)	0.47	4.8	14.0

Table S17. Flexibility dividends when the secondary-skill efficiency is drawn per technician and secondary trade instead of being one shared number. Dividends are in weighted tardiness units against the dedicated baseline of the same crew multiplier, on the seven-rule-and-Random envelope and under fixed EDD; $\bar{\eta}$ is the realized mean of the draw. A capture ratio is printed only where the full-flexibility dividend clears the 2% evaluability threshold, and a dash marks the cells where it does not. The first row of each panel is the scalar penalty of the main grid. $n = 763$ instances per cell.

		Rules envelope					Fixed EDD	
Draw	$\bar{\eta}$	CHAIN	GEN	FULL	FULL/L0	ρ	CHAIN	FULL
A. $m = 0.6$, dedicated baseline 328.97								
Scalar $\eta = 0.8$	0.800	4.11	4.72	3.21	1.0%	–	6.46	1.81
Per technician $U[0.60, 1.00]$								
seed 20260901	0.800	3.66	2.72	0.69	0.2%	–	6.01	−0.57
seed 20260902	0.801	2.76	1.59	2.17	0.7%	–	5.11	−2.40
seed 20260903	0.797	2.92	3.05	1.19	0.4%	–	5.28	0.32
pooled	0.800	3.11	2.45	1.35	0.4%	–	5.47	−0.88
Per technician $U[0.80, 1.00]$								
seed 20260901	0.900	9.04	9.09	8.54	2.6%	1.06	11.39	9.88
seed 20260902	0.901	8.62	8.45	8.54	2.6%	1.01	10.98	9.28
seed 20260903	0.899	8.84	9.27	8.52	2.6%	1.04	11.20	10.17
pooled	0.900	8.84	8.94	8.53	2.6%	1.04	11.19	9.78
B. $m = 0.8$, dedicated baseline 318.45								
Scalar $\eta = 0.8$	0.800	0.09	−0.12	−0.05	0.0%	–	0.09	−1.73
Per technician $U[0.60, 1.00]$								
seed 20260901	0.799	−0.99	−0.76	−0.21	−0.1%	–	−0.99	−3.39
seed 20260902	0.800	−0.17	0.30	−0.33	−0.1%	–	−0.17	−2.96
seed 20260903	0.800	0.44	0.61	−0.31	−0.1%	–	0.44	−1.89
pooled	0.800	−0.24	0.05	−0.28	−0.1%	–	−0.24	−2.75
Per technician $U[0.80, 1.00]$								
seed 20260901	0.899	1.79	2.06	2.27	0.7%	–	1.79	1.39
seed 20260902	0.900	1.84	1.93	2.30	0.7%	–	1.84	1.45
seed 20260903	0.900	2.24	2.24	2.05	0.6%	–	2.24	1.49
pooled	0.900	1.96	2.08	2.21	0.7%	–	1.96	1.44

Table S18. Both pre-specified verdicts re-read at neighboring settings of the three numerical thresholds, from the released results. Panel A varies the capture-ratio threshold at $m = 0.6$ and $\eta = 1.0$, the family that carries the chaining verdict. It is read against the point estimate and against the lower end of the 95% instance-cluster interval, on the rules-and-policy envelope and under fixed EDD. Panel B varies the evaluability threshold. Each column head carries that family's full-flexibility dividend as a share of its dedicated baseline. The $m = 0.7$ family is post hoc, and the last column is the penalized family the chaining reading is quoted in. Panel C varies the number of policy seeds Gate P requires to beat the best ranked rule, alongside the probability that at least that many of ten fair coin flips come up ahead. $n = 763$ instances per cell, and the policy pool is the ten verdict-class seeds.

A. Capture-ratio threshold ($m = 0.6$, $\eta = 1.0$)				
Threshold	Envelope 0.870	CI floor 0.794	Fixed EDD 0.874	CI floor 0.781
0.60	met	met	met	met
0.70 (released)	met	met	met	met
0.80	met	not met	met	not met
B. Evaluability threshold; the first three columns are at $\eta = 1.0$				
Threshold	$m = 0.6$ (4.4%)	$m = 0.7$ (3.2%)	$m = 0.8$ (1.3%)	$m = 0.6$, $\eta = 0.8$ (1.0%)
1%	evaluable	evaluable	evaluable	not evaluable
2% (released)	evaluable	evaluable	not evaluable	not evaluable
3%	evaluable	evaluable	not evaluable	not evaluable
5%	not evaluable	not evaluable	not evaluable	not evaluable
C. Seeds Gate P requires ahead of the best ranked rule (1 of 10 are ahead in every scope)				
Seeds required	Pooled	$m = 0.8$	$m = 0.6$	Coin-flip P
6 of 10	not met	not met	not met	0.377
7 of 10	not met	not met	not met	0.172
8 of 10 (released)	not met	not met	not met	0.055
9 of 10	not met	not met	not met	0.011

Hypothetical Crew Sizes for Long Orders

The work-order records hold no per-order technician count, so the main model reads the logged labor hours p_j of an order as the uninterrupted work of one technician. The crew-size analysis replaces this reading for long orders with hypothetical crews. These crews are assumed scenarios and do not describe observed practice. An order longer than 8 bh needs

$$c_j = \min\left(c_{\max},\ \lceil p_j/8 \rceil,\ k_g\right)$$

technicians at once, where k_g is the primary headcount of the order's trade g and the cap $c_{\max}$ takes the values 1, 2, 3 and 4. Shorter orders keep $c_j = 1$, and $c_{\max} = 1$ is the main model. The cap k_g ensures that a trade's own primary technicians can always form the crew, so a long order of a trade with a single primary technician stays single. At $m = 0.6$, 97.8% of the long orders become crew orders under every cap $c_{\max} \geq 2$. All members of a crew start together and stay busy until the order ends. The elapsed time is p_j over the crew's combined rate, rounded up to the 0.01 bh grid, and the combined rate counts 1 for a primary-skill member and η for a secondary-skill member. Every order therefore keeps its labor content up to that rounding.

Dispatch keeps the released rules and adds reservation. At each decision instant the rule ranks the waiting orders by its unchanged score. If at least c_j eligible technicians are idle, the highest-ranked order starts with c_j of them, chosen by the released technician tie-break (Text S4). Otherwise the dispatcher holds those idle technicians for the order until the next event, and the rule continues with the remaining orders and technicians. Holds are decided again at every event, which lets a crew assemble as its members finish other work. Panel (a) of Table S20 reports the cost of this reservation as longer waits for crew orders and as technician-hours held idle. Rules that read processing time score a crew order by p_j/c_j, its elapsed time when every member works at the primary rate. EDD and pFIFO never

Table S19. Static reference resolved by instance size (E1′, $m = 1.0$): model size, proof rates, wall to proof, best-rule optimality gaps, and rule latency. $\sum_j |M_j|$ counts the optional intervals of the CP-SAT model and is the mean over the instances of the row. Wall to proof covers the proved instances only, and adds the 60 s run to the 300 s re-run where the tail supplied the proof. Because it is measured to the certificate over proved instances, it differs from the median 60 s run wall of Table S11, which is taken over every instance of the cell. The optimality gap is the best ranked rule's mean distance above the certified optimum on the certified subset; the bound gap is its mean distance above the strongest non-zero solver lower bound. A dash marks a row with no certified instance. Latency is the median per-decision wall of the ranked rules. Solver walls are as released, at two CP-SAT search workers per solve. $n = 120$ instances per row.

		proved		wall to proof (s), median / p90		best-rule gap		rule
Λ (η)	$\sum_j \lvert M_j \rvert$	60 s	300 s	median	p90	optimum	bound	ms/dec.
50 orders								
L0	1,149	100%	100%	0.2	0.5	0.00%	0.00%	0.016
CHAIN(1.0) (1.0)	2,149	100%	100%	0.3	0.8	0.00%	0.00%	0.019
CHAIN(1.0) (0.8)	2,149	100%	100%	1.1	2.4	0.00%	0.00%	0.019
FULL (1.0)	6,938	100%	100%	1.7	3.9	0.00%	0.00%	0.043
FULL (0.8)	6,938	100%	100%	11.5	24.2	0.00%	0.00%	0.043
150 orders								
L0	3,467	100%	100%	0.8	2.2	0.00%	0.00%	0.017
CHAIN(1.0) (1.0)	6,437	100%	100%	1.7	4.4	0.00%	0.00%	0.034
CHAIN(1.0) (0.8)	6,437	100%	100%	8.3	22.2	0.00%	0.00%	0.034
FULL (1.0)	20,812	100%	100%	8.5	15.3	0.00%	0.00%	0.065
FULL (0.8)	20,812	7%	91%	165.8	239.4	0.00%	0.00%	0.065
400 orders								
L0	9,308	99%	99%	3.6	7.4	0.16%	0.17%	0.029
CHAIN(1.0) (1.0)	17,325	99%	99%	7.2	12.5	0.03%	0.02%	0.033
CHAIN(1.0) (0.8)	17,325	49%	99%	104.9	178.1	1.80%	0.52%	0.033
FULL (1.0)	55,500	98%	100%	32.7	52.7	0.02%	0.01%	0.064
FULL (0.8)	55,500	0%	0%	–	–	–	0.00%	0.059

Note: at 400 orders under FULL (0.8) no instance carries a non-zero lower bound. Its bound-gap entry therefore covers only the 47 instances whose best rule already reaches zero tardiness and is optimal for that reason; on the other 73 the distance from the optimum is unknown.

read processing time, so the fixed-EDD estimate does not depend on this choice.

Two checks verify the crew engine. With every crew of one, it reproduces the released weighted tardiness exactly on all 91,560 rule rows of the Tier-1 grid, and the makespan, mean flow time, and breach shares match as well. An independent validator, which imports neither the crew engine nor the released engine, checks every crew schedule. It verifies crew size, the eligibility of every member, a shared start and end, the elapsed time, the release time, and the absence of overlap, and it recomputes the weighted tardiness. Only the seven ranked rules and Random run in this analysis, because the learned policies were trained on one-technician orders and never saw a crew order.

A post-protocol decomposition by a length floor explains why crews change the absolute scale of the dividends. An order that would end after its due date even if it started at release is late in every schedule. The sum $\sum_j w_j (r_j + e_j - d_j)^+$, with e_j the order's shortest elapsed time under the crew rule, is therefore a lower bound on the weighted tardiness of every schedule. No crew design and no dispatcher can remove this floor. The L0 tardiness above it is tardiness caused by waiting for a technician. In the one-technician model at $m = 0.6$, the floor is 314.4 of the 329.0 weighted units of L0 tardiness on the rules envelope (95.6%). Full flexibility removes 99.7% of the remaining waiting-induced tardiness, and the chain removes 85.1%. The dividend is therefore small because the removable part of the tardiness is small. Crews shorten long orders, so the floor falls to 86.1, 37.7, and 20.4 units for $c_{\max} = 2$, 3, and 4 at $m = 0.6$. The waiting-induced part grows to 19.5, 24.0, and 25.4 units, and Δ(FULL) grows with it to 18.4–65.6% of L0 across the two estimators.

These larger dividends make the $m = 0.8$ and $m = 1.0$ families evaluable as well. Across the 9 crew families at $\eta = 1.0$ ($c_{\max} \geq 2$), the chain captures 0.79–0.88 of the full-flexibility dividend under fixed EDD and 0.73–0.88 on the

Table S20. Capture ratio under hypothetical crews for orders longer than 8 bh (crew rule and reservation dispatch in Text S8). The crew-size cap $c_{\max}$ limits the technicians an order may need at once, and $c_{\max} = 1$ is the one-technician model of the main manuscript. Panel (a) describes the crew orders and the L0 schedules of the rule that sets the rules-envelope L0 level, including the mean wait from release to start and the technician-hours held idle per instance while crews assemble. In panels (b) and (c), L0, the dividends, and Δ(FULL) as a percentage of L0 are rules-envelope values. A dash in a ρ column marks a family below the 2% evaluability threshold on that estimator, and a shown ρ carries its 95% instance-cluster bootstrap interval. $n = 763$ instances per cell.

(a) Crews and L0 schedules

		Crew orders		L0 TWT on	Mean wait (bh)		Held (bh)
m	$c_{\max}$	Share	Size	orders > 8 bh	Crew	Single	per instance
0.6	1	–	–	98.7%	–	2.00	0.0
	2	9.1%	2.00	94.7%	3.56	2.40	8.8
	3	9.1%	2.47	88.1%	4.29	2.56	19.2
	4	9.1%	2.74	80.6%	4.64	2.61	23.8
0.8	1	–	–	99.7%	–	1.61	0.0
	2	9.3%	2.00	98.2%	2.18	1.33	5.0
	3	9.3%	2.46	94.8%	2.69	1.45	13.1
	4	9.3%	2.74	90.7%	2.91	1.50	17.2
1.0	1	–	–	99.9%	–	1.13	0.0
	2	9.3%	2.00	99.1%	1.35	1.40	2.6
	3	9.3%	2.46	97.9%	1.73	1.54	8.2
	4	9.3%	2.75	95.6%	1.94	1.02	11.0

(b) Capture ratio, $\eta = 1.0$

m	$c_{\max}$	L0	Δ(CHAIN)	Δ(FULL)	% of L0	ρ (EDD)	ρ (envelope)
0.6	1	329.0	12.43	14.57	4.4%	0.87 (0.78–0.93)	0.85 (0.78–0.91)
	2	105.6	15.73	19.44	18.4%	0.85 (0.76–0.91)	0.81 (0.74–0.87)
	3	61.8	18.93	23.92	38.7%	0.85 (0.78–0.91)	0.79 (0.72–0.86)
	4	45.9	18.53	25.23	55.0%	0.83 (0.70–0.91)	0.73 (0.65–0.83)
0.8	1	318.5	3.42	4.08	1.3%	–	–
	2	92.8	5.71	6.86	7.4%	0.84 (0.73–0.93)	0.83 (0.73–0.92)
	3	47.5	8.48	9.92	20.9%	0.87 (0.79–0.93)	0.86 (0.79–0.92)
	4	31.2	9.30	11.21	35.9%	0.86 (0.78–0.92)	0.83 (0.76–0.89)
1.0	1	315.5	0.85	1.16	0.4%	–	–
	2	89.2	2.95	3.34	3.7%	0.88 (0.80–0.97)	0.88 (0.80–0.97)
	3	40.5	3.28	3.93	9.7%	0.84 (0.74–0.92)	0.84 (0.74–0.92)
	4	23.7	3.62	4.62	19.5%	0.79 (0.68–0.89)	0.78 (0.67–0.88)

(c) Capture ratio, $\eta = 0.8$

m	$c_{\max}$	L0	Δ(CHAIN)	Δ(FULL)	% of L0	ρ (EDD)	ρ (envelope)
0.6	1	329.0	4.11	3.21	1.0%	–	–
	2	105.6	10.65	8.13	7.7%	1.19 (1.05–1.61)	1.31 (1.08–1.81)
	3	61.8	14.66	14.93	24.2%	0.99 (0.92–1.07)	0.98 (0.87–1.11)
	4	45.9	14.35	17.23	37.5%	0.91 (0.77–1.00)	0.83 (0.68–1.00)
0.8	1	318.5	0.09	−0.05	0.0%	–	–
	2	92.8	2.94	2.60	2.8%	1.30 (−0.41–4.88)	1.13 (0.64–2.94)
	3	47.5	6.06	5.65	11.9%	1.06 (0.94–1.24)	1.07 (0.93–1.27)
	4	31.2	7.16	7.24	23.2%	0.99 (0.91–1.09)	0.99 (0.89–1.11)
1.0	1	315.5	−0.59	−0.70	−0.2%	–	–
	2	89.2	1.29	1.83	2.1%	–	0.70 (0.14–1.29)
	3	40.5	1.98	1.99	4.9%	1.02 (0.79–1.72)	0.99 (0.69–1.33)
	4	23.7	2.46	2.55	10.8%	0.97 (0.80–1.24)	0.96 (0.72–1.22)

rules envelope. The envelope ratio falls as crews grow, and at $m = 0.6$ with $c_{\max} = 4$ its interval reaches below the 0.70 threshold (0.734 (95% CI 0.654–0.832)). At $\eta = 0.8$ the ratio spans 0.70–1.31 over the 17 evaluable estimates, and it exceeds 1 in 7 of them, where the chain outperforms full flexibility. The 87% capture ratio of the main manuscript is therefore the one-technician value. Under every tested crew reading, the chain still clears the 0.70 threshold by point estimate at $\eta = 1.0$.

Frame U at Matched Aggregate Offered Load

Frame U compares the structures on generator-drawn workloads at a fixed aggregate offered-load utilization U, the window's labor hours over headcount times the 80 bh window. Within one U level, each instance holds three inputs constant across L0, CHAIN(1.0), and FULL. These inputs are the order stream (release times, trades, labor hours, priorities, and due dates), the 80 bh window, and the headcount of every trade at the calibrated crew sizes ($m = 1.0$). The realized aggregate load is therefore identical across the three structures. Eligibility varies, and with it the capacity that each trade's queue can draw on, the dispatch decisions, and the method that the envelope selects. At $\eta = 0.8$, every secondary-skill assignment also takes longer. Each U level draws its own 30 instances per campus, so the instances differ between U levels.

Table S21. Offered load by trade in Frame U under the dedicated roster. A trade's load is its labor hours over its headcount times the 80 bh window, so a load above 1 exceeds the trade's own capacity. Each value is the mean over the 120 instances of a U level, and the last two columns give the shares of trades, and of all labor hours, in trades loaded above 1.

U	Aggregate load	Mean over trades	Hours-weighted mean	Most loaded trade	Trades above 1	Hours in trades above 1
0.9	0.90	0.78	0.97	1.28	29%	52%
1.0	1.00	0.86	1.09	1.43	37%	64%
1.1	1.10	0.96	1.20	1.56	45%	76%
1.3	1.30	1.13	1.40	1.83	60%	86%

Table S22. Weighted tardiness in Frame U at matched aggregate offered load, as pooled mean TWT per instance with the paired chain-minus-full difference and the capture ratio against the dedicated roster. Panel (a) uses the best observed method and panel (b) fixed EDD. Intervals are 95% bootstrap intervals over instances resampled within campus and U. In panel (a) the best method is re-selected in every resample. A dash marks a family below the 2% evaluability threshold, and no value is set in bold. $n = 120$ instances per U level and 360 pooled.

η	U	L0	CHAIN(1.0)	FULL	CHAIN − FULL (95% CI)	% of FULL	Δ(FULL) over L0	ρ (95% CI)	Evaluable
(a) Best observed among the seven ranked rules and both policy classes									
1.0	0.9	2912	2886	2866	20 (17–23)	0.7%	1.6%	–	No
	1.0	3368	3337	3303	34 (30–40)	1.0%	1.9%	–	No
	1.1	3774	3710	3662	48 (44–53)	1.3%	3.0%	0.57 (0.48–0.64)	Yes
	1.3	4391	4235	4155	79 (73–86)	1.9%	5.4%	0.66 (0.61–0.71)	Yes
	≥1.0, pooled	3844	3761	3707	54 (51–57)	1.5%	3.6%	0.61 (0.56–0.65)	Yes
0.8	0.9	2912	3034	3331	−297 (−339 to −247)	−8.9%	−14.4%	–	No
	1.0	3368	3549	4112	−563 (−606 to −518)	−13.7%	−22.1%	–	No
	1.1	3774	4037	4755	−719 (−757 to −680)	−15.1%	−26.0%	–	No
	1.3	4391	4655	5546	−891 (−937 to −843)	−16.1%	−26.3%	–	No
	≥1.0, pooled	3844	4080	4805	−724 (−749 to −699)	−15.1%	−25.0%	–	No
(b) Fixed EDD (no method selection)									
1.0	0.9	2912	2887	2866	21 (18–24)	0.7%	1.6%	–	No
	1.0	3368	3337	3303	34 (30–40)	1.0%	1.9%	–	No
	1.1	3774	3710	3662	48 (44–53)	1.3%	3.0%	0.57 (0.48–0.64)	Yes
	1.3	4391	4235	4155	79 (73–86)	1.9%	5.4%	0.66 (0.61–0.71)	Yes
	≥1.0, pooled	3844	3761	3707	54 (51–57)	1.5%	3.6%	0.61 (0.56–0.65)	Yes
0.8	0.9	2912	3034	3331	−297 (−348 to −249)	−8.9%	−14.4%	–	No
	1.0	3368	3549	4139	−591 (−643 to −537)	−14.3%	−22.9%	–	No
	1.1	3774	4037	4769	−733 (−770 to −697)	−15.4%	−26.4%	–	No
	1.3	4391	4655	5573	−918 (−966 to −872)	−16.5%	−26.9%	–	No
	≥1.0, pooled	3844	4080	4827	−747 (−774 to −721)	−15.5%	−25.6%	–	No

Equal aggregate load does not give equal effective capacity, because trade loads differ widely under the dedicated roster (Table S21). At $U = 1.0$, 37% of the trades, holding 64% of the labor hours, are loaded above their own capacity, and the most loaded trade averages a load of 1.43. Full flexibility pools capacity across these trades, which the dedicated

roster cannot do. Frame U is therefore a matched-aggregate-load comparison, in which trade-level bottlenecks still differ between the structures.

Table S22 reports two readings of the same runs. The first is the pre-specified fallback for the $m = 0.8$ family, which compares the chain with full flexibility directly at $U \geq 1.0$. At $\eta = 1.0$, the chain's pooled TWT on the best observed method exceeds that of full flexibility by 1.45% (95% CI 1.37–1.54) of FULL's TWT. This interval excludes zero and lies entirely between 1% and 2%, so the table reports the difference as measured proximity and claims no equivalence. At $\eta = 0.8$, the chain-minus-full difference is −15.08% (95% CI −15.46 to −14.68) of FULL's TWT, so the chain outperforms full flexibility once secondary work is penalized. The second reading anchors both structures to the dedicated roster and is a post-protocol, descriptive analysis. At $\eta = 1.0$, the chain captures 0.61 (95% CI 0.56–0.65) of the dedicated-to-full reduction pooled over $U \geq 1.0$, with 0.57 (95% CI 0.48–0.64) at $U = 1.1$ and 0.66 (95% CI 0.61–0.71) at $U = 1.3$. These estimates lie below the 0.70 threshold, and at $U = 1.0$ the full-flexibility dividend of 1.94% of L0 falls below the 2% evaluability threshold. At $\eta = 0.8$, both flexible structures exceed the dedicated roster's TWT at every U, so no capture ratio exists there.

A post-protocol split of the same runs under fixed EDD applies the length-floor decomposition of the crew-size analysis above. Pooled over $U \geq 1.0$ at $\eta = 1.0$, the floor is 96.3% of L0 tardiness, against 94.9% in the replayed weeks at $m = 0.6$ under fixed EDD. Full flexibility removes 96.0% of the waiting-induced remainder, close to the 99.7% it removes in the replayed weeks. The part of that remainder that the chain removes is 58.4%, against 87.1% in the replayed weeks. The lower capture ratio in Frame U therefore comes from a smaller chain share of the waiting that full flexibility removes. The two frames differ in how they generate work and create contention. Frame D replays the recorded weeks of each campus and reduces headcount with the crew multiplier m, while Frame U keeps the calibrated headcount ($m = 1.0$) and raises load through the arrival rate. In Frame U, each trade's orders arrive from an independent Poisson process at a rate fitted on the training years, with lognormal labor hours fitted per trade. This comparison does not identify which of these differences lowers the chain's share.

References

Zhang, Z. and Ku Chia, T. P. (2026). "When does learned dispatching beat priority rules? An open benchmark for technician-constrained maintenance work-order scheduling in building portfolios." SSRN preprint, DOI 10.2139/ssrn.7095162. Code and benchmark at https://github.com/NTUZZH/FM-Scheduling.